Department of Computer Science
BCS - 2A
Lab Mid (Version 1)
All questions are mapped to CLO4

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A vehicle rental company wants to develop a Java application to manage vehicles available for rent. Each vehicle has a rental agreement containing the rental duration, daily rental rate, and customer's name. The company manages regular vehicles and premium vehicles. A premium vehicle has an additional luxury category and receives a 15% discount on the calculated rental cost.

Implement the following requirements.

- **Part A — (5 marks)**
  1. Create a RentalAgreement class containing customerName (String), rentalDays (int), dailyRate (double). Initialize data members and also include a method calculateRentalCost() that returns the rental cost before any discount.

  2. Create a Vehicle class containing vehicleId (int), vehicleModel (String), rentalAgreement (RentalAgreement). Initializes all attributes. Include a method displayDetails() to display vehicle and rental agreement information.
- **Part B — (4 marks)**
  3. Create a PremiumVehicle class that inherits from Vehicle. Add an attribute luxuryCategory (String). Use super to initialize inherited attributes. Override calculateTotalCost() to apply a 15% discount to the rental cost. Override displayDetails() to display vehicle details, the luxury category, and the final rental cost. In the Vehicle class, implement calculateTotalCost() to return the normal rental cost.
- **Part C — (5 marks)**
  4. In a class named RentalManager, implement the following methods:
    - void updateDailyRate(RentalAgreement r, double newRate): Update the daily rate of an existing rental agreement, provided the new rate is positive. The daily rate must be updated in the existing RentalAgreement object, not by creating a replacement object.
    - void displayVehicle(Vehicle v): Accept a vehicle object and display its details by calling displayDetails().
- **Part D — (6 marks)**
  5. In RentalManager, implement Vehicle findVehicleWithHighestRentalCost(Vehicle[] vehicles)
  This method must receive an array of Vehicle objects. Find and return the vehicle with the highest final rental cost, including any applicable premium discount. Return null if the array is empty or contains no vehicle objects. If multiple vehicles have the same final cost, return the first one encountered.
- **Part E — (5 marks)**
  6. Write the main() method to:
    - Create an array capable of storing five Vehicle objects.
    - Store at least three Vehicle objects and two PremiumVehicle objects.
    - Display all vehicles using RentalManager.
    - Update one vehicle's daily rate by passing its existing RentalAgreement object to updateDailyRate().
    - Display the updated vehicle details.
    - Call findVehicleWithHighestRentalCost() and display the returned vehicle's model and final rental cost.

Department of Computer Science
BCS - 2A
Lab Mid (Version 2)
All questions are mapped to CLO4

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A university maintains a system for lending equipment to students. Each equipment item has a lending record containing the borrower's name, number of borrowing days, and daily rental charge. The university lends regular equipment and specialized equipment. Specialized equipment has an additional category and receives a 20% discount on the total lending charge.

Implement the following requirements.

Part A — (5 marks)

1. Create an LendingRecord class containing borrowerName (String), borrowingDays (int),dailyCharge (double). Initialize all the attributes and add a method calculateCharge() that returns borrowingDays × dailyCharge.

2. Create an Equipment class containing: equipmentId (int), equipmentName (String) lendingRecord (LendingRecord). Include a method displayDetails() to display equipment and lending information, and a method calculateTotalCharge() that returns the normal lending charge.

Part B — (4 marks)

3. Create a SpecializedEquipment class that is type of Equipment. Add an attribute equipmentCategory (String). Initialize all attributes including inherited. Override calculateTotalCharge() to apply a 20% discount to the normal charge. Override displayDetails() to display the equipment details, category, and final charge.

Part C — (5 marks)

4. In a class named EquipmentManager, implement:

- void updateDailyCharge(LendingRecord r, double newCharge): Update the daily charge of an existing lending record if the new charge is positive. The update must modify the existing LendingRecord object.
- void displayEquipment(Equipment e): Accept an equipment object and display its details by calling displayDetails().

Part D — (6 marks)

5. Implement the following method in EquipmentManager
Equipment findMostExpensiveEquipment(Equipment[] items)
The method must search the array for the equipment with the highest final charge, including the discount for specialized equipment. Return null if the array is empty or contains no equipment objects and return the first matching object if multiple items have the same final charge.

Part E — (5 marks)

6. Write a main() method that:

- Creates an array capable of storing five Equipment objects.
- Stores three Equipment objects and two SpecializedEquipment objects.
- Displays all equipment using EquipmentManager.
- Updates the daily charge of one item's lending record by passing the existing object to updateDailyCharge().
- Displays the updated equipment details.
- Calls findMostExpensiveEquipment() and displays the equipment name and final charge.